WellnexUS

UCS503 Software Engineering Project Report

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1. PROJECT SELECTION PHASE

1.1 Software Bid

UCS 503- Software Engineering Lab

Group no. 3Q14 Dated: 29/08/2025

Team Name: Synergy Squad
Team Members: Aryan Tyagi, Harnoor Kaur, Kalpesh Paliwal, Krishna Agarwal, Sparsh Bhaskar

Name	Roll No	Project Experience	Programming Language used	Signature
Aryan Tyagi	102317088	Movie Recommendation System: Developed an intelligent, AI-driven Movie Recommendation System that enhances user experience by providing personalized movie suggestions using KNN model and cosine similarity.	Python, Jupyter Notebook, Scikit-learn, Pandas, KNN, Matplotlib HTML, CSS, JS, React	
Harnoor Kaur	102317099	Advanced ML-Based Heart Disease Predictor: Built a ML model using the UCI dataset with preprocessing (missing value handling), normalization and encoding. Trained and evaluated multiple classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost etc.) using metrics like accuracy, F1score, confusion matrices and ROC-AUC.	Python, Jupyter Notebook, Scikit-learn, Pandas, Numpy, Seaborn, Matplotlib	
Kalpesh Paliwal	102317087	Airbnb Full-Stacked Project: Built an Airbnb-style web app with Node.js, Express, MongoDB, and EJS (MVC), featuring listings, reviews, multistep booking, Mapbox integration, and a Geminipowered AI chatbot for real-time user assistance.	Javascript,MongoDb, Express.js, Node.js ,React	
Krishna Agarwal	102317102	Hotel Management System: Developed a web-based application that manages room availability, reservations, bookings, and provides a customer feedback portal with a user-friendly frontend and secure backend.	HTML, CSS, JavaScript, Node.js, React	

Sparsh Bhaskar	102317013	**ETMS (Employee Task Management System)** | Engineered a full-stack web application integrating secure user authentication, role-based access, and RESTful API endpoints for task data exchange. Utilized Django ORM for efficient data handling and implemented CRUD functionalities for complete task lifecycle management	Python, Django, SQLite, HTML/CSS	

Programming Language / Environment Experience

- JavaScript (React.js, Node.js, Express.js)
- Python (Flask, TensorFlow, Keras)
- MongoDB
- HTML/CSS/TailwindCSS

Choices of Projects:

	Project Name	Unique Selling Point
First Choice	**WellnexUS –** **Motivation:** Many people struggle to maintain physical and mental wellbeing due to lack of personalized guidance and empathetic support. WellnexUS addresses this by enabling users to track calories, receive personalized exercise and nutrition plans, and interact with a supportive mental wellness chatbot, making holistic health management simple and engaging. **Reason for Choice:** The project leverages strengths in full stack development, AI/ML for recommendations and chatbot functionality, and modern UI design. It is highly relevant to today's health-conscious culture, with gender-based personalization ensuring tailored physiological and psychological guidance for improved effectiveness and engagement. **Feasibility Analysis:** Implementation is feasible using Python for AI/ML models, TensorFlow/Keras for advanced recommendation or sentiment analysis, React for interactive dashboards and chat interfaces, and Node.js/Flask with MongoDB Atlas for	Revolutionary digital platform that integrates physical and mental wellbeing management in one place. Users can set personalized goals, track calories and nutrients, receive exercise and lifestyle recommendations tailored to their gender and objectives, and engage with an AI-powered mental health chatbot via text and voice for instant support. The platform provides dynamic progress tracking, adaptive feedback, and motivational interaction, all delivered through an intuitive, responsive design—making holistic self-care easy, personalized, and accessible for everyone.

	scalable backend data storage and user management.	
Second Choice	**CookSense AI –** **Motivation:** Many people struggle to decide what to cook with the ingredients they already have. CookSense AI solves this problem by making meal planning effortless and creative. **Reason for Choice:** The idea aligns with our strengths in AI/ML, image recognition, and responsive web development. It's also highly relevant in today's fast-paced lifestyle where convenience and personalization are valued. **Feasibility Analysis:** The project can be implemented using Python (for AI/ML), TensorFlow/Keras (for image recognition), and a responsive frontend framework like React.	Revolutionary web platform combining traditional recipe browsing with AI-powered ingredient recognition. Users can photograph available ingredients to get instant recipe suggestions, browse curated recipes with ratings, save favorites, and follow step-bystep cooking instructions with builtin timers. Responsive design with user comments and nutritional information.
Third Choice	**JustiFy –** **Motivation:** Many common people in India struggle to understand legal documents, processes, and jargon. Access to affordable legal advice is also limited. *Digital Legal Assistant* solves this by simplifying legal knowledge, summarizing complex documents, and guiding users in plain English/Hindi. **Reason for Choice:** The idea aligns with growing needs in legal accessibility, AI/NLP integration, and web development. It's socially impactful, empowering rural and urban users alike with affordable, instant legal assistance. **Feasibility Analysis:** The project can be implemented using Python/Node.js for backend, React for frontend, MongoDB/MySQL for data, Tesseract for OCR, and LLMs (OpenAI/Gemini) for document summarization and chatbot guidance.	A revolutionary web platform providing simplified legal help to common people through AIpowered document summarization, legal chatbot guidance, and readymade document templates. Features include OCR for scanned legal files, voice input, legal dictionary, user feedback, and optional lawyer connect, with responsive design for easy use in rural and urban areas
Fourth Choice	**MediConnect+** **Motivation:** Access to quality healthcare remains a challenge, especially in rural areas. MediConnect+ bridges this gap by providing telemedicine services, secure health record storage, and AI-assisted health insights.	All-in-one telemedicine platform combining AI-powered symptom checks, virtual consultations, and secure health record management. Provides 24/7 access to healthcare from any location. Ensures patient privacy with encrypted, userfriendly data storage.

	Reason for Choice: It leverages our knowledge in AI, web development, and secure data handling while addressing a real-world need with social impact potential. **Feasibility Analysis:** We can implement the project using secure backend frameworks (Node.js, Django), cloudbased encrypted storage, AI APIs for symptom checking, and WebRTC for video consultations.	

Additional Remarks:

Our team selected WellnexUS due to its significant social impact potential and alignment with current health technology trends. The project leverages our strengths in full-stack development, AI/ML integration, and modern UI/UX design. We're passionate about creating an accessible solution that addresses the growing need for integrated physical and mental health management. As a team, we feel that this project is particularly convincing because it goes beyond typical fitness trackers by integrating both physical health and mental wellbeing into one platform. The inclusion of a supportive chatbot and gender-based personalization makes it socially impactful and user-centric. With rising stress levels, sedentary lifestyles, and increasing demand for accessible digital health solutions, WellnexUS has strong real-world relevance and potential adoption. Moreover, working on this project offers the opportunity to combine skills in AI/ML, full stack development, and human-centered design to create a meaningful solution that can genuinely improve people's quality of life.

1.2 Project Overview

1.2.1 Purpose

The primary purpose of **WellnexUS** is to provide users with a unified, intuitive platform for managing their overall wellness—including fitness, nutrition, mental health, and lifestyle habits. By simplifying the wellness-tracking process, WellnexUS empowers individuals, students, working professionals, and fitness enthusiasts to monitor and improve their physical and mental wellbeing through personalized insights and actionable guidance. The platform focuses on enhancing user experience through smart tracking tools, AI-driven recommendations, and customized wellness plans, ensuring that users can build sustainable habits and achieve their health goals effectively.

We've carefully considered the diverse range of users who will engage with this system, so this document outlines the different user groups and explains how the interface and features are designed to meet their needs. You'll also find the hardware and software requirements necessary to operate the platform smoothly.

Since different stakeholders—such as end-users, administrators, wellness experts, and the development team—have unique expectations from WellnexUS, we have ensured that all perspectives are addressed. This section gives a clear understanding of what the system aims to achieve, what functionalities it supports, and how it is envisioned to function in real-world use cases.

1.2.2 Scope of the Develpoment Project

WellnexUS will offer a comprehensive digital wellness platform that enables users to manage and improve their overall health through integrated tools for **fitness tracking, nutrition management, and mental wellness support**. The project includes a user-friendly interface where individuals can log their meals, monitor physical activity, track emotional states, and receive AI-driven recommendations tailored to their personal health goals.

The system will provide features such as guided workout plans, mood tracking, meditation sessions, and access to curated wellness content. It will also support importing health data from connected devices, real-time analytics, and progress visualization through charts and dashboards. Additionally, the platform will include community challenges, and secure data management services. Users will be able to download wellness reports and share insights with trainers, doctors, or wellness experts as needed.

Overall, the scope of the WellnexUS project covers end-to-end functionality for creating a personalized wellness experience that promotes healthier habits and sustained lifestyle improvements.

1.2.3 Intended Audience

The intended audience for **WellnexUS** includes individuals of all age groups who seek to improve their overall health and wellbeing through an integrated platform. This encompasses fitness enthusiasts, individuals managing lifestyle conditions, nutrition-conscious users, and those seeking mental wellness support.
Additionally, **health coaches, fitness trainers, therapists, and wellness organizations** can leverage the platform to track client progress, provide personalized guidance, and host wellness programs. **Educational institutions and corporate wellness programs** may also adopt WellnexUS as a resource for promoting balanced lifestyles among students and employees.

The software must be able to perform the following operations:

1. **User Registration and Authentication:**
    - Users should be able to create secure accounts using email or social logins (Google, Apple).
    - A multi-step onboarding should collect basic health details such as age, gender, fitness goals, and preferences.
2. **Profile and Health Data Management:**
    - Users should have the ability to input and update personal information, including physical details, activity levels, and medical conditions.
    - Options to update passwords, upload profile pictures, and manage notification settings should be available.
3. **Goal Setting and Tracking:**
    - Users should be able to set wellness goals (e.g., weight loss, mental balance, fitness improvement).
    - Progress tracking through graphs, charts, and summaries should be provided for daily, weekly, and monthly views.
4. **Nutrition Management:**
    - A food logging system should allow users to record daily meals with calories and nutrient values.
    - Integration with food databases and barcode scanners should be available for quick input.
5. **Workout and Activity Tracking:**
    - Users should be able to select from AI-recommended workout plans or create custom sessions.
    - Real-time workout tracking with performance summaries should be supported.

6. **Mental Wellness and AI Coaching:**
    - An interactive AI wellness coach should assist users with meditation, stress management, and mood analysis.
    - Daily motivational prompts and mental health check-ins should be available.
7. **Community and Social Interaction:**
    - Users should be able to join wellness challenges, share progress, and interact with others through forums or leaderboards.
    - Experts can host group workshops and Q&A sessions for users.
8. **Analytics and Reports:**
    - The system should generate detailed health and wellness reports based on user activity, diet, and mood trends.
    - Insights should be presented through an intuitive dashboard for easy interpretation.

1.2.4 Definitions, Abbreviations and Acronyms

Definitions

1. **Authentication:**
   Authentication is the process of verifying the identity of a user before allowing access to the WellnexUS platform. Users provide valid login credentials (email, password, or social login) to ensure secure access to their wellness data and personal information.
2. **Registration:**
   Registration refers to the process of creating a new user account on WellnexUS. Users must provide essential information, such as name, email address, age, gender, and password to set up a unique profile on the platform.
3. **User Profile:**
   The user profile contains personal and health-related details such as physical measurements, medical conditions, activity levels, dietary preferences, and wellness goals. This information is used to generate personalized recommendations across nutrition, fitness, and mental wellness.
4. **Goal Setting:**
   Goal setting involves defining user objectives such as weight loss, muscle gain, step count, calorie intake, hydration targets, or mental wellness milestones. These goals guide the overall progress tracking within the application.
5. **Nutrition Tracking:**
   This refers to the process of logging daily meals, calories, and nutrients. The system uses food databases, barcode scanning, and manual entry to help users record and analyze their dietary habits.

6. **Workout Session:**
   A workout session is a scheduled or completed fitness activity tracked by the system. Users can follow AI-generated workout plans or create custom routines, with details stored for performance analysis.

7. **AI Coach / Wellness Assistant:**
   The AI Coach is a smart assistant that provides personalized guidance in areas like nutrition, workouts, mental health, and daily motivation. It analyses user data and generates tailored recommendations based on patterns and goals.

8. **Mood Tracking:**
   Mood tracking allows users to record feelings, emotions, and mental states throughout the day. This enables WellnexUS to generate insights linking mood with diet, sleep, activity, and stress levels.

9. **Meditation & Mindfulness:**
   This refers to guided mental wellness exercises such as meditation sessions, breathing routines, and stress-relief practices offered within the platform.

10. **Challenge Participation:**
    Challenges are community or individual wellness events (e.g., step challenge, hydration challenge). Users join challenges, track progress, and compete or collaborate with others.

11. **Analytics & Insights:**
    This refers to personalized health reports generated by WellnexUS, including trends for calories, moods, workouts, hydration, and progress toward goals.

12. **Community Engagement:**
    Users can participate in groups, forums, or discussions to share progress, access expert advice, and interact with others for motivation.

Abbreviations and Acronyms

Acronym	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BMI	Body Mass Index
DB	Database
FAQ	Frequently Asked Questions
GPS	Global Positioning System

Acronym	Full Form
GUI	Graphical User Interface
MFA	Multi-Factor Authentication
OTP	One-Time Password
PB	Personal Best
PIN	Personal Identification Number
SRS	Software Requirements Specification
SQL	Structured Query Language
UI	User Interface
UX	User Experience

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1.3 Feasibility Report

1.3.1 Product Perspective

1. **User Registration:**
    - Visit the WellnexUS website or mobile application.
    - Click on the **“Sign Up”** button.
    - Provide your email address, create a password, and enter basic details such as name, age, gender, and fitness goal.
2. **User Login:**
    - After completing registration, log in using your email and password.
    - Optionally, use social login options like Google or Apple for quick access.
3. **User Profile:**
    - Upon logging in, you will land on your personalized profile dashboard.
    - Here, you can update personal details including height, weight, medical conditions, dietary preferences, and profile picture.
4. **Setting Wellness Goals:**
    - Click on **“Set Goals”** to define health objectives such as calorie targets, step count, hydration goals, weight goals, or mental wellness preferences.
    - Goals can be modified anytime based on your progress.
5. **Nutrition Tracking:**
    - Navigate to the **“Nutrition”** section to log your meals.
    - Choose from options like manual entry, barcode scanning, food database search, or voice input.
    - The system automatically calculates calories and nutrients for each meal.
6. **Workout Planning:**
    - Click on **“Workouts”** to access AI-generated routines or browse the exercise library.
    - Select a workout template and begin your session, tracking calories burned and duration.
7. **Mental Wellness Tools:**
    - Open the **"Mental Health”** section to access meditation sessions, mood tracking tools, and AI-driven emotional support chat.
    - Record mood entries and receive insights based on sleep, diet, and activity.

8. **Community Challenges:**
    - Join wellness challenges such as step goals, hydration streaks, or workout competitions.
    - Track your rank, earn badges, and interact with other participants.
9. **Analytics and Progress Reports:**
    - Visit the **"Analytics"** tab to view progress charts for calories, steps, workouts, hydration, and mood trends.
    - Compare weekly and monthly performance through interactive graphs.
11. **Downloading Reports:**
    - Users can download detailed health and progress reports in formats like PDF for personal records or consultation with healthcare professionals.
    - Reports include nutrition summaries, mood trends, workout metrics, and recommendations.

1.3.2 Product Functions

1.User Registration and Profiles

• User Registration:

- Users should be able to register using their email address or social media accounts (e.g., Google, Apple).
- A secure verification process should confirm the user's identity before account activation.

• User Profiles:

- Each user should have a profile page where they can manage personal details such as age, gender, height, weight, medical conditions, and preferences.
- Users should be able to upload a profile picture and update contact information when needed.
- Health and wellness preferences should be customizable for personalized recommendations.

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2. Nutrition Tracking and Management

• Meal Logging:

- Users should be able to log daily meals through multiple methods such as search, barcode scanning, voice input, or manual entry.
- Each meal should display nutritional information including calories, carbs, protein, and fats.

• Meal Management:

- Users should be able to edit, delete, or duplicate meal entries easily.
- The system should calculate total daily intake and compare it with user goals.
- Users can maintain meal history for long-term tracking.

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3. Workout Planning and Activity Tracking

• Workout Creation:

- Users should be able to start guided workouts using customizable workout templates.
- Each workout should include sections such as warm-up, exercises, reps, sets, and cooldown.

• Workout Editing:

- Users should be able to modify, reschedule, or repeat previous workout sessions.
- Users should be allowed to save multiple workout versions tailored for different goals (e.g., fat loss, muscle gain).

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4. Template Selection and Browsing (Workout & Meal Plans)

• Template Library:

- Users should be able to browse a library of professionally designed workout and diet templates.
- Each template should show details like difficulty, duration, expected results, and required equipment.

• Search and Filtering:

- Users should be able to search for templates using keywords (e.g., “weight loss”, “yoga”, “high protein”).

- Advanced filtering options such as duration, difficulty level, goal type, and diet preference should be available.

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5. Mental Wellness and AI Interaction

• Mood Tracking:

- Users should be able to record moods with labels (e.g., happy, stressed, tired) and intensity levels.
- Mood history should be accessible for long-term mental wellness insights.

• AI Chat & Mindfulness:

- Users can interact with the AI Wellness Coach for emotional support, guidance, and daily motivation.
- The system should offer meditation sessions, breathing exercises, and stress-relief practices.

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6. Community Challenges and Social Engagement

• Challenge Participation:

- Users should be able to join wellness challenges such as step challenges, hydration streaks, or workout streaks.
- Challenge leaderboards should display rankings among participants.

• Social Interaction:

- Users should be able to share progress, achievements, and milestones within community groups.
- Users may follow other members or experts for motivation and guidance.

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7. Analytics and Insights

• Progress Tracking:

- Users should be able to view detailed analytics for calories, steps, workouts, hydration, and mood.
- Weekly, monthly, and custom time range insights should be available.

• **Reports:**

- Users should be able to download detailed wellness reports for consultation or personal records.
- AI-generated recommendations should accompany analytics for better decision-making.

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8. Admin Panel

• **Admin Management:**

- An admin panel should be available for managing user accounts, wellness content, challenges, and issue reports.

• **User Moderation:**

- Admins should have the ability to remove, suspend, or restrict users violating platform policies.

• **Template Oversight:**

- Admins should be able to add, modify, or remove workout templates, meal templates, and wellness resources to maintain quality.

1.3.3 User Characteristics

1. General Users:

- The primary user base consists of individuals focused on improving their overall wellness, including fitness enthusiasts, students, working professionals, and adults aged 16–45 who want to track health, nutrition, workouts, and mental wellbeing.
- These users typically engage in logging daily meals, participating in workouts, tracking moods, joining community challenges, and interacting with the AI Wellness Coach for guidance. They value an intuitive interface, personalized recommendations, and easy-to-understand analytics.

2. Wellness Seekers (Goal-Oriented Users):

- This group includes users actively pursuing specific goals such as weight loss, muscle gain, stress reduction, or healthier lifestyle habits. They often check progress charts, follow recommended plans, and adjust workout or diet routines based on insights.
- They expect detailed tracking tools, motivational reminders, structured plans, and a smooth user experience across all wellness modules.

3. Administrators:

- Administrators ensure smooth platform operations, maintain system health, and uphold community standards. They manage user accounts, monitor activities, review reported issues, and maintain the quality of wellness content.
- Admins also analyze user feedback, update templates, and make data-driven decisions to enhance functionality, security, and overall user satisfaction.

1.3.4 General Constraints, Assumptions and Dependencies

1. Constraints:

• Budget:

The available development budget will determine the scope and scale of the WellnexUS platform. Limited funds may require prioritizing essential features such as authentication, wellness tracking, and AI chatbot over advanced analytics or premium enhancements.

• Time:

Project deadlines or defined launch dates may impose time constraints. Development, testing, and deployment schedules must be realistic to ensure the platform is stable, secure, and fully functional at launch.

• Technology Stack:

The selection of programming languages, frameworks, cloud services, and AI models may be constrained by team expertise, licensing costs, and existing infrastructure. Integrations like nutrition APIs, AI models, and fitness tracking services may require compatible technologies.

• Data Privacy & Compliance:

The platform must comply with Indian data protection norms, security standards, and health-related privacy considerations, which can limit design choices and data handling methods.

• Device Compatibility:

The platform must operate smoothly on mobile phones, tablets, and desktops, which may restrict UI design complexity and performance-heavy features.

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2. Assumptions and Dependencies:

1. User Proficiency:

- It is assumed that users accessing WellnexUS have basic digital literacy and can navigate mobile or web applications without difficulty.
- Users are expected to understand basic wellness terms like calories, steps, hydration, etc.

2. User Authentication:

- It is assumed that every user will access the platform through a valid email ID or social login (Google/Apple).
- Users will safeguard their credentials and are responsible for maintaining account security.

3. Administrative Oversight:

- It is assumed that dedicated administrators will be available to monitor user activities, manage content, handle reports, and maintain database integrity.
- Admins will oversee premium services, expert content, challenges, and AI model updates.

4. Internet Connectivity:

- It is assumed that users will have access to stable internet connectivity to log workouts, meals, and interact with the AI Wellness Coach.
- Low connectivity may limit access to features such as live chat or analytics.

5. Software Requirements:

- It is assumed that users will have a modern web browser or the WellnexUS mobile app installed on their devices.
- Notifications, real-time tracking, and chat features may depend on device permissions and OS compatibility.

6. Wellness Content and Template Availability:

- The availability of high-quality workout templates, diet charts, and meditation programs may depend on partnerships with fitness trainers, nutritionists, and wellness experts.

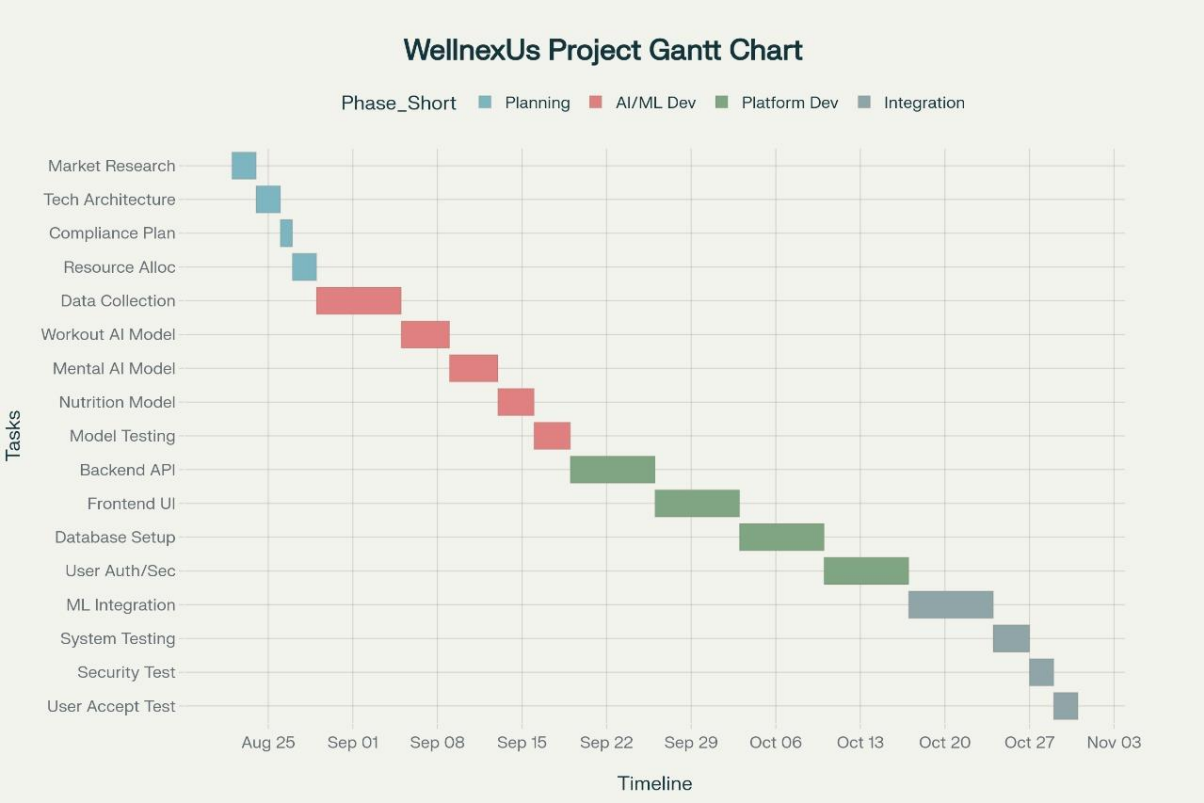
7. Third-Party Integrations:

- The platform may depend on external services such as nutrition databases, AI APIs, and fitness tracking APIs.
- Any downtime or policy changes from these services may impact WellnexUS functionality.

1.4 Gantt Chart

The Gantt chart for the WellnexUS project illustrates the complete development lifecycle and dependencies across major phases of the wellness platform. The project begins with the **Planning Phase**, which includes market research, technical architecture design, compliance checks, and resource allocation to establish a solid foundation. This is followed by the **AI/ML Development Phase**, where datasets are collected and multiple machine learning models—such as workout recommendation, mental wellness analysis, and nutrition prediction—are trained and validated. Once the AI components are prepared, the project transitions into the **Platform Development Phase**, covering backend API creation, frontend UI development, database setup, and user authentication and security implementation. These tasks run in a structured sequence to ensure smooth integration. After platform components are built, the **Integration Phase** merges AI models with the application, followed by system testing, security testing, and user acceptance testing to ensure reliability and performance.

Overall, the Gantt chart clearly visualizes task durations, start and end dates, and interdependencies across all modules, enabling efficient progress tracking and ensuring the WellnexUS platform is delivered on time with high quality.



Wellness-US Project Timeline Overview

Phase 1: Planning (Aug 25 – Aug 31)
This phase is about defining what the project needs and making sure the foundation is solid.

- **Market Research:** Understand the wellness app market and user needs.
- **Tech Architecture:** Decide the system design, tools, and backend/frontend stack.
- **Compliance Plan:** Ensure regulatory standards (e.g., privacy, medical guidance rules).
- **Resource Allocation:** Assign team roles, sprint cycles, and infrastructure.

Phase 2: AI / ML Development (Sep 1 – Sep 20)
Core intelligence of the platform is created here.

- **Data Collection:** Acquire and preprocess wellness datasets.
- **Workout AI Model:** Build AI that suggests workouts based on user profile.
- **Mental Health AI Model:** Model for stress, mindfulness, and mood tracking.
- **Nutrition Model:** Diet recommendation engine.
- **Model Testing:** Validate accuracy, remove bias, optimize performance.

Phase 3: Platform Development (Sep 20 – Oct 18)
The app and backend platform get developed in this stage.

- **Backend API:** Create data access, AI interaction endpoints, and business logic.
- **Frontend UI:** Build the actual user-facing app screens and interactions.
- **Database Setup:** Deploy, structure, and optimize user + model output storage.
- **User Authentication & Security:** Login system, encryption, secure session management.

Phase 4: Integration & Testing (Oct 18 – Nov 3)
Everything comes together for final polishing and readiness.

- **ML Integration:** Connect AI models to the live app.
- **System Testing:** Check performance, responsiveness, workflow reliability.
- **Security Testing:** Ensure no vulnerabilities or data leaks.
- **User Acceptance Testing (UAT):** Real users test the app before official release.

Total Duration:

~10 Weeks from **Aug 25** to **Nov 3**

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2. ANALYSIS PHASE

2.1 Use Case Diagram

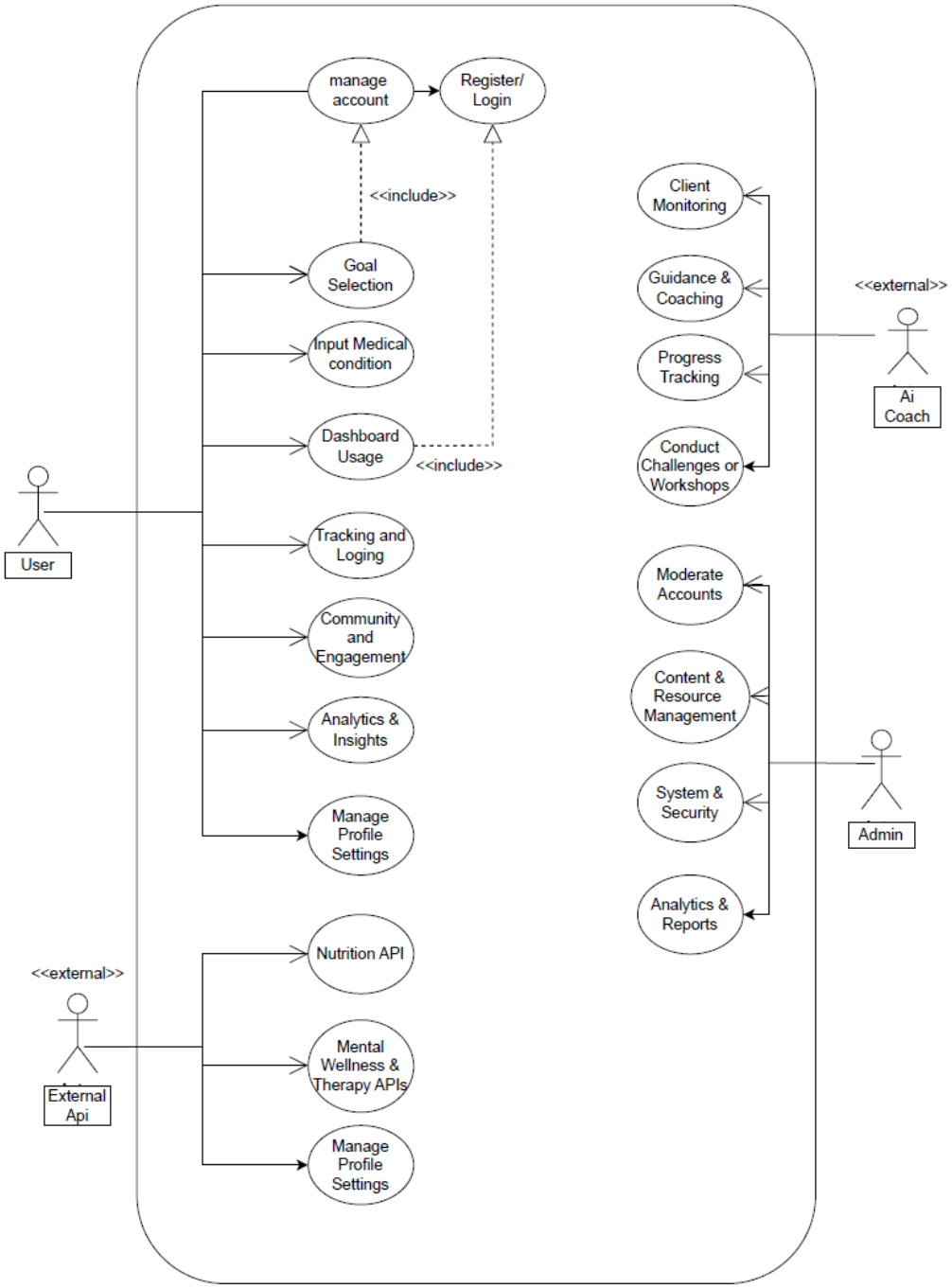


Diagram Description:

Actors

- **User:**
  Primary actor who registers or logs in, selects goals, inputs medical conditions, tracks and logs daily activity, views analytics, and manages personal profile settings.

- **AI Coach:**
  External intelligent component that monitors client progress, offers personalized guidance and coaching, tracks performance, and conducts virtual challenges or workshops.
- **Admin:**
  System administrator responsible for moderating accounts, managing content and resources, maintaining security, and generating analytical reports.
- **External API:**
  Third-party services that supply nutrition information (Nutrition API) and mental-wellness resources (Therapy APIs).

Relationships

- **<<include>>** dependencies are used for common functionality such as authentication and dashboard access:
  - *Manage Account* includes *Register/Login*.
  - *Goal Selection* and *Tracking and Logging* include *Dashboard Usage*.
- **<<extend>>** relationships represent optional or premium functions such as AI-driven coaching and workshops.
- **Generalization** may apply to different user types (e.g., Basic User, Premium User) or API providers.

2.2 Use-Case Templates

Use Case 1: User Registration and Health Profile Setup

Attribute	Details
Use Case Title	User Registration and Health Profile Setup
Use Case ID	UC-101
Actors	New User
Description	A new user creates an account and completes comprehensive health profile setup
Preconditions	User has access to WellnexUS application, not currently logged in
Task Sequence	1. User selects "Sign Up" 2. Enters email and password 3. Verifies email address 4. Completes health questionnaire (age, gender, weight, height, goals) 5. Sets activity level and preferences 6. System generates initial health metrics and goals
Postconditions	User account created, personalized dashboard available, initial recommendations generated
Priority	High
Authors	Team WellnexUS

Use Case 2: Daily Nutrition Tracking

Attribute	Details
Use Case Title	Daily Nutrition Tracking
Use Case ID	UC-102
Actors	Registered User
Description	User logs food consumption throughout the day using multiple input methods
Preconditions	User is logged in, has completed profile setup
Task Sequence	1. User navigates to Nutrition section 2. Selects meal type (breakfast/lunch/dinner/snack) 3. Chooses input method (search, barcode scan, voice) 4. Enters/confirms food items and quantities 5. System calculates and displays nutritional information 6. User reviews and saves entry
Postconditions	Meal logged, daily nutrition totals updated, progress metrics recalculated
Priority	High
Authors	Team WellnexUS

Use Case 3: AI Chatbot Mental Wellness Session

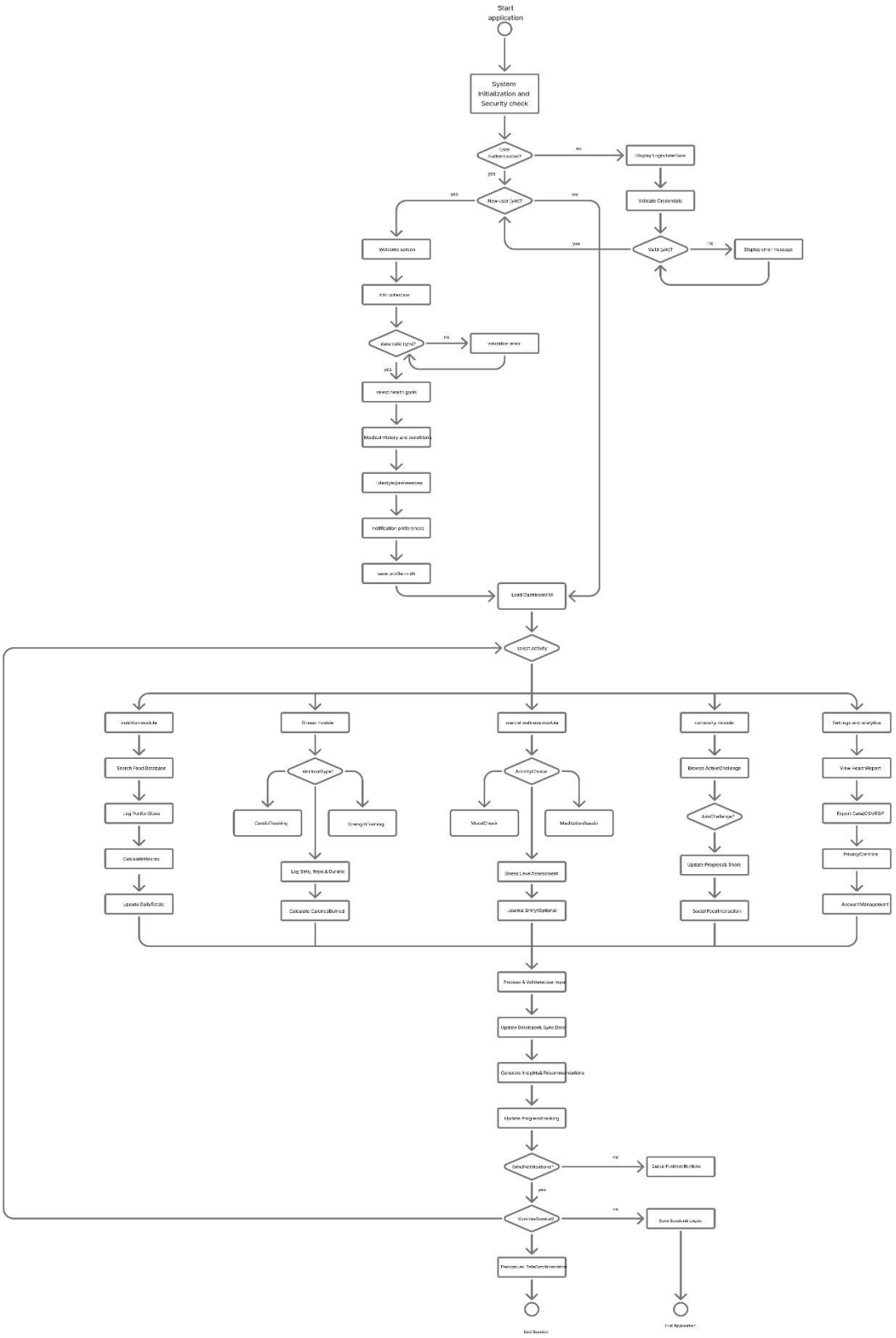
Attribute	Details
Use Case Title	AI Chatbot Mental Wellness Session
Use Case ID	UC-103
Actors	Registered User, AI System
Description	User engages in conversational mental health support session with AI chatbot
Preconditions	User is logged in, chatbot service available
Task Sequence	1. User navigates to Mental Wellness section 2. Selects "Chat with Wellness Assistant" 3. Chooses conversation mode (text/voice) 4. Engages in dialogue with AI chatbot 5. System provides empathetic responses and coping strategies 6. Session summary and recommendations provided
Postconditions	Chat history saved, mood insights generated, follow-up actions suggested
Priority	Medium
Authors	Team WellnexUS

Use Case 4: Personalized Workout Generation

Attribute	Details
Use Case Title	Personalized Workout Generation
Use Case ID	UC-104
Actors	Registered User, AI System
Description	System generates customized workout plan based on user profile and progress
Preconditions	User profile complete, workout history available
Task Sequence	1. User requests new workout plan 2. System analyzes user data (goals, progress, preferences) 3. AI generates personalized exercise routine 4. User reviews and customizes plan 5. System provides exercise demonstrations and instructions 6. Plan saved to user calendar
Postconditions	New workout plan created, calendar updated, progress tracking initialized
Priority	High
Authors	Team WellnexUS

2.3 Activity Diagram and Swimlane Diagram

1.Activity Diagram

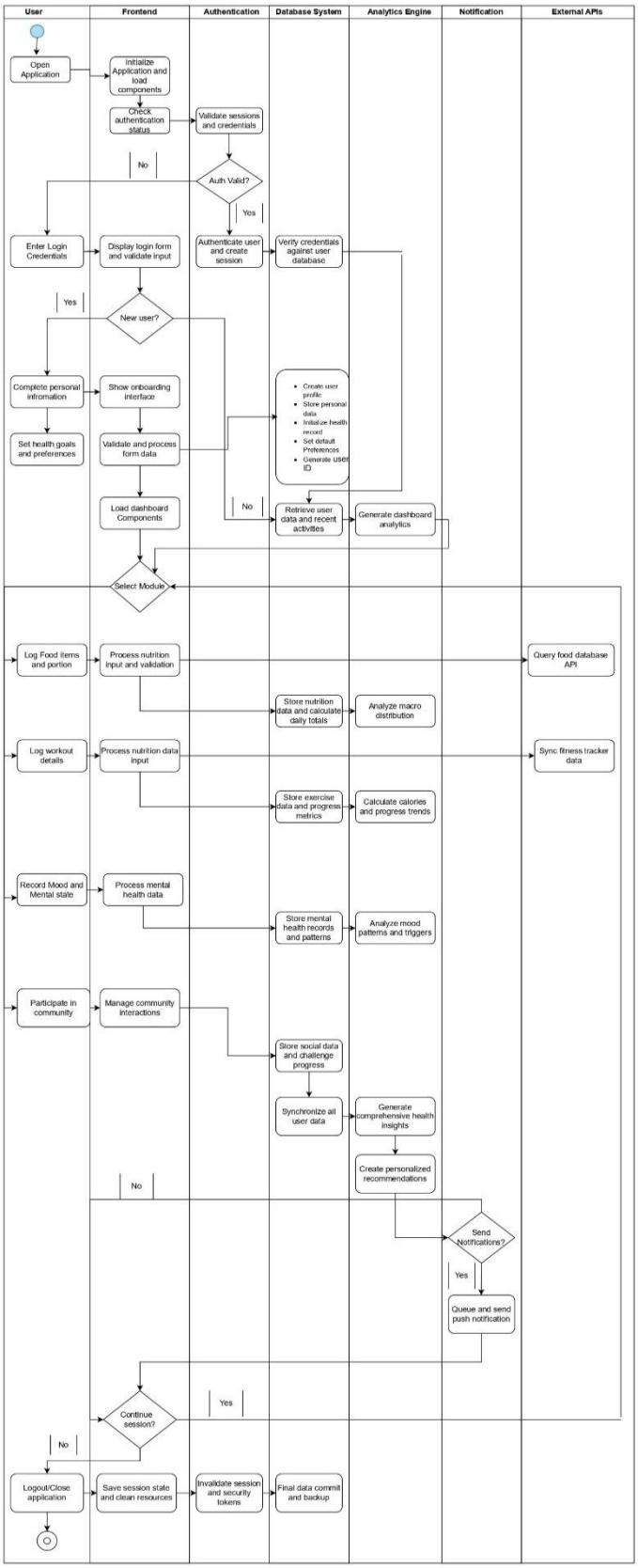


Activity Diagram Description:

Start → System Initialization & Security Check → Decision: New User / Existing User

- **New User Path:** Register → Set goals → Input medical details → Personalize dashboard → Load main interface
- **Existing User Path:** Login → Access personalized dashboard
  Within the dashboard, the user can:
  • **Nutrition Tracking:** Browse food database → Log meals → View calorie summary → Update daily progress
  • **Workout Tracking:** View plan → Start workout → Log completion → Update fitness metrics
  • **Mental Wellness:** Check mood → Access therapy or meditation → Log reflection → Update emotional insights
- End → Generate analytics & feedback → Save session → Logout

2.Swimlane Diagram



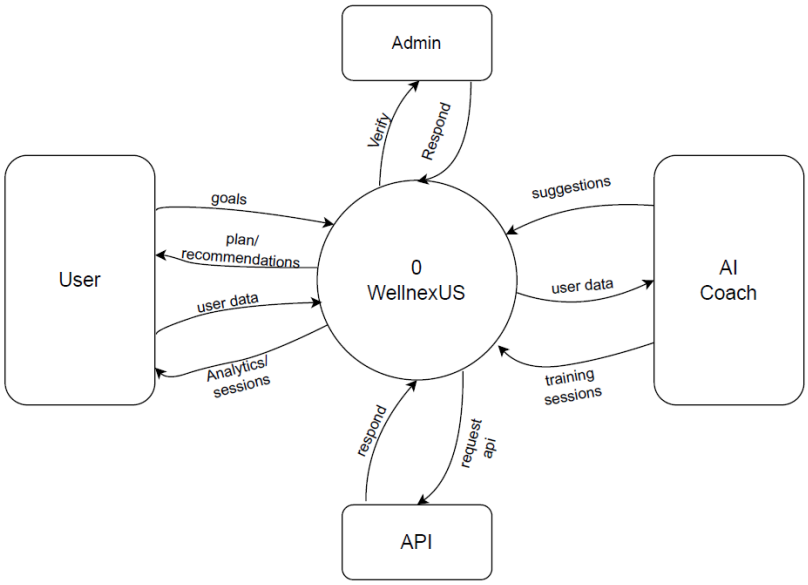
Swimlane Diagram Description:

The swimlane diagram outlines the coordinated workflow across multiple components of the **WellnexUS** system:

- **User Lane:** Opens the app → Logs in or registers → Inputs personal, nutritional, and wellness data → Views dashboard and results.
- **Frontend Lane:** Initializes application → Displays login and onboarding interfaces → Processes user input and module selections.
- **Authentication Lane:** Validates sessions and credentials → Authenticates users → Manages tokens and sessions.
- **Database System Lane:** Stores user data, nutrition logs, workouts, and mood entries → Retrieves historical records.
- **Analytics Engine Lane:** Analyzes data for trends → Generates personalized insights and health recommendations.
- **Notification Lane:** Sends alerts and reminders based on analytics or goal progress.
- **External APIs Lane:** Syncs with nutrition, fitness, and mental health APIs for external data enrichment.

2.4 Data Flow Diagram

2.4.1 DFD Level-0



DFD Level 0 (Context Diagram):

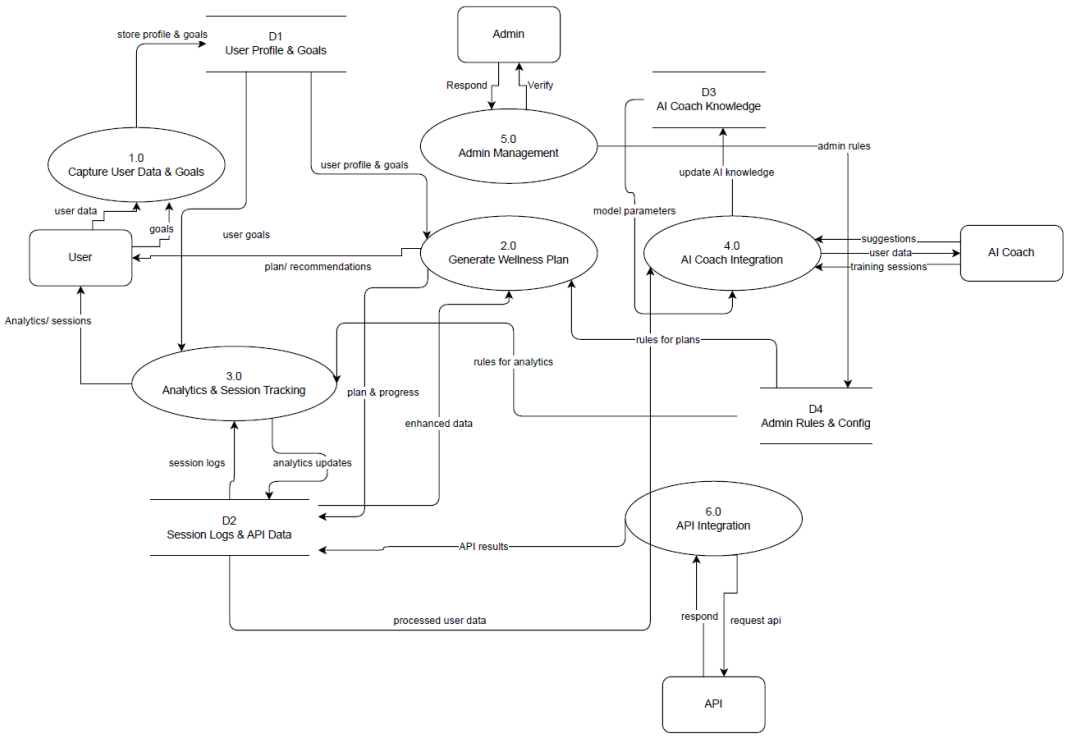
 User: Sends personal goals and health data → Receives personalized plans, recommendations, and analytics reports.

 Admin: Verifies user data, manages system access, and receives analytical feedback from the system.

 AI Coach: Exchanges user data and delivers tailored workout, nutrition, and wellness suggestions.

 External API: Provides third-party data such as nutrition databases and training content through API requests and responses.

2.4.2 DFD Level-1

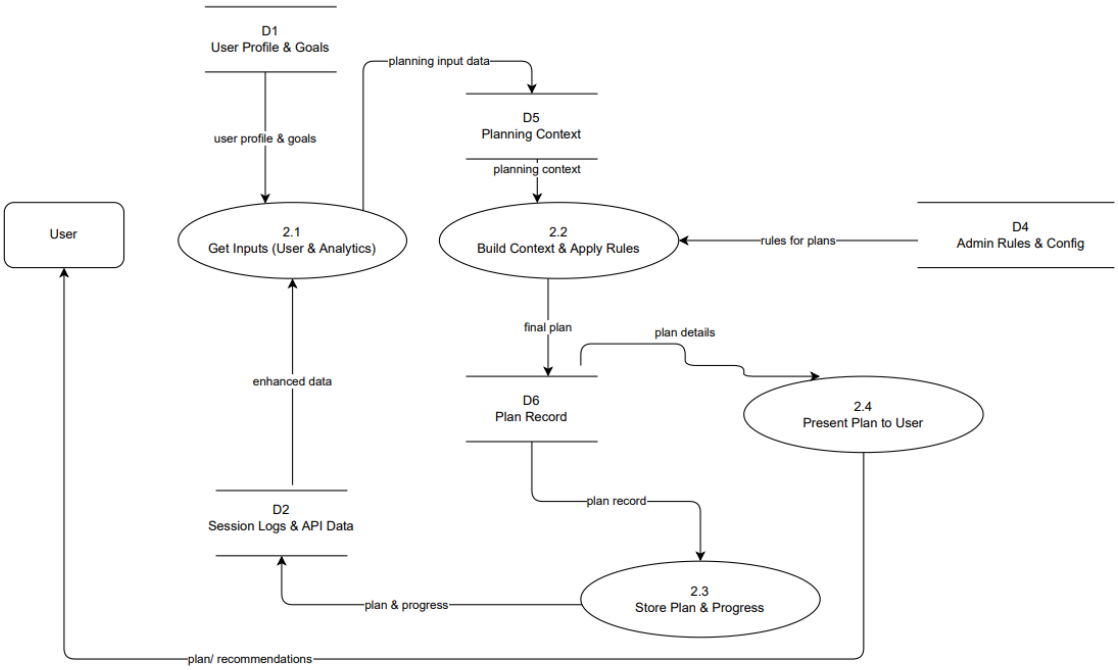


DFD Level-1 Description (Wellness-US System)

The Level-1 Data Flow Diagram illustrates the major functional processes of the Wellness-US platform and the flow of data among users, admin, data stores, and external systems. The system begins by capturing user data and wellness goals, which are stored in the *User Profile & Goals (D1)* repository. Based on this information, the system generates personalized wellness plans (Process 2.0). User interactions, progress, and session logs are continuously tracked through *Analytics & Session Tracking (3.0)* and stored in the *Session Logs & API Data (D2)* repository.

The *AI Coach Integration (4.0)* utilizes analytics data and stored knowledge to provide personalized suggestions and interactive guidance to users. Admins manage system rules, model parameters, and AI training updates via *Admin Management (5.0)*, which updates both AI knowledge (D3) and rules/configuration (D4). Additionally, the *API Integration (6.0)* facilitates external communication with third-party services, enhancing user data processing and system intelligence.

Overall, the Level-1 DFD shows how user data flows through various functional units to generate wellness plans, track progress, provide AI-powered coaching, and enable admin supervision to deliver a dynamic and personalized wellness experience.



DFD Level-2 Description – Generate Wellness Plan (Process 2.0)

This shows the internal decomposition of how user health information and analytics are processed to create a personalized wellness plan in WellnexUS. This level expands the single Level-1 process into four detailed subprocesses and two internal data stores.

2.1 Get Inputs (User & Analytics)

This process retrieves the user's stored profile and goal information from D1 – User Profile & Goals, along with historical session and API-based recommendations from D2 – Session Logs & API Data. The collected inputs are forwarded internally to the planning system.

2.2 Build Context & Apply Rules

The retrieved data is combined and structured into the D5 – Planning Context store. Administrative rules for plan generation, fetched from D4 – Admin Rules & Config, are then applied to produce a personalized wellness plan. The resulting plan is stored in D6 – Plan Record.

2.3 Store Plan & Progress

This process updates historical analytics by saving the generated plan and progress indicators into D2 – Session Logs & API Data, ensuring future recommendations are more accurate.

2.4 Present Plan to User

The finalized wellness plan details from D6 – Plan Record are formatted and delivered to the user as personalized recommendations and wellness guidance.

Internal Data Stores:

D5 – Planning Context: Temporary storage for combined user and analytics data during plan creation.

D6 – Plan Record: Stores finalized plans and progress details for future reference and analytics.

2.5 Software Requirement Specification

2.5.1 User Interfaces

1. Home Page Interface:

- The home page provides an intuitive and welcoming layout featuring clear call-to-action buttons such as **"Log In"**, **"Sign Up"**, and **"Start Your Wellness Journey."**
- Users can quickly access key features like **Nutrition Tracking**, **Workout Plans**, **AI Wellness Chat**, and **Daily Tips**. A search bar allows users to look up workouts, food items, or wellness resources instantly.
- Quick links to motivational articles, health tips, and featured wellness challenges help users begin exploring the platform.

2. Login Page Interface:

- The login page allows users to sign in using their email, password, or social login options such as **Google** or **Apple**.
- A clean input form is presented, with validation prompts for incorrect entries.
- A **"Forgot Password?"** option helps users reset their password securely through email or OTP verification.

3. Wellness Dashboard Interface:

- After logging in, users land on their personalized dashboard displaying daily calorie intake, activity summary, hydration levels, mood status, and wellness recommendations.
- Interactive widgets allow users to log meals, start a workout, track their mood, or chat with the AI Coach with one click.
- Graphs and progress rings offer a snapshot of weekly and monthly performance.

4. Nutrition Tracking Interface:

- This interface allows users to log meals using multiple input options such as search, barcode scanning, manual entry, or voice input.
- Users can view detailed nutrition information for each food item, including calories, macronutrients, and portion sizes.
- The interface also suggests healthier alternatives, recipe ideas, and meal templates based on user preferences.

5. Workout Interface:

- Users can browse workout categories such as strength training, yoga, HIIT, and cardio.
- Each workout plan displays difficulty level, duration, required equipment, and demo videos.
- Users can start a guided workout session featuring timers, rep counters, and real-time progress tracking.

6. Mental Wellness Interface:

- This interface includes tools for mood tracking, meditation sessions, breathing exercises, and AI-driven emotional support.
- Users select mood icons, write journal entries, and view emotional trends over time.
- Meditation sessions include calming visuals, background sounds, and timers for guided practice.

7. Template Selection Interface (Workout & Diet Plans):

- The interface presents libraries of curated templates for **workout routines**, **meal plans**, and **wellness schedules**.
- Users can filter templates by fitness goals, diet preferences, difficulty level, and duration.
- Each template includes a preview mode for users to review before applying it to their plan.

8. Profile Management Interface:

- Users can update personal information such as name, age, height, weight, medical conditions, and wellness preferences.
- The interface allows uploading or changing profile pictures, managing connected devices, and adjusting notification settings.
- Users can view their activity history, downloaded reports, , and linked health data.

2.5.2 Functional Requirements

1. User Registration and Authentication

• Description:
Users must be able to create an account using their email or social login options. The system should securely validate user credentials and maintain secure sessions.

• Requirements:

- Users should receive a verification email or OTP upon registration.
- Users must be able to reset their passwords securely.
- Social login options (Google/Apple) should be available.
- The system must prevent duplicate accounts using the same email.

---

2. User Profile Management

• **Description:**
Users should have a personal profile page where they can manage their wellness information, body metrics, preferences, and linked devices.

• **Requirements:**

- Users can edit personal details (name, age, gender, contact info, profile picture).
- Users can update health details (height, weight, medical conditions, diet preferences).
- Users can manage connected devices (fitness bands, smart watches).
- Users can adjust notification and privacy settings.

---

3. Nutrition Tracking and Meal Management

• **Description:**
Users should be able to log meals, view their nutritional breakdown, and track their calorie intake throughout the day.

• **Requirements:**

- Users can add meals using search, barcode scanning, voice input, or manual entry.
- The system should calculate calories, carbs, protein, and fats automatically.
- Users can edit, delete, or duplicate previously logged meals.
- Users can view daily and weekly nutrition summaries.

---

4. Workout Management

• **Description:**
Users should be able to start workouts, follow guided routines, and track exercise performance.

• **Requirements:**

- Users can choose from an exercise library with images/videos.
- Users can follow pre-made workout templates or customized plans.
- Users can track reps, sets, duration, and calories burned.
- Users can save completed workout sessions for history and analytics.

---

5. Mental Wellness and Mood Tracking

• **Description:**
Users should have tools for emotional wellness, including mood logging, AI chat support, and guided mindfulness sessions.

• **Requirements:**

- Users can log their mood with emotion labels and intensity.
- Users can start guided meditation or breathing sessions.
- The AI Wellness Coach should provide supportive, context-based responses.
- Users should be able to view their emotional history and trends.

---

6. AI Personal Recommendations

• **Description:**
The system should generate personalized wellness recommendations using user data and AI models.

• **Requirements:**

- Recommendations should be based on nutrition, workouts, mood, and activity levels.
- Users should receive daily suggestions such as meal ideas, workouts, or hydration reminders.
- The system should update recommendations dynamically as user behavior changes.

---

7. Community Challenges & Social Engagement

• **Description:**
Users must be able to participate in wellness challenges and interact within the community.

• **Requirements:**

- Users can join challenges (steps, calories, hydration, workouts).
- Users can view their rank on leaderboards.
- Users can share achievements with the community.
- The system should reward users with badges or points.

---

8. Analytics and Reporting

• **Description:**
Users must be able to view detailed insights about their wellness activities.

• **Requirements:**

- Users can access daily, weekly, and monthly charts.
- The system should provide trend analysis for meals, workouts, mood, and habits.
- Users can download progress reports in PDF format.
- AI should highlight patterns or warning signs (e.g., low mood or inactivity).

---

9. Admin Panel & Platform Management

• **Description:**
Admins should be able to oversee users, content, system health, and reported issues.

• **Requirements:**

- Admins can manage user accounts (block, suspend, edit).
- Admins can add/edit/remove workout templates, meal data, and wellness content.
- Admins can monitor community challenges and review reports.
- Admins can view system analytics and performance logs.

2.5.3 Non-Functional Requirements

1. Performance

• **Description:**
The platform should respond quickly, process wellness data efficiently, and support simultaneous user activity without performance degradation.

• **Requirements:**

- Dashboards, logs, and analytics pages should load within **2 seconds** under normal conditions.
- The system should support at least **5,000 concurrent users** during peak health-tracking hours.
- AI responses (chatbot + recommendations) should be generated within **3 seconds**.

---

2. Security

• **Description:**
User health data must be protected using advanced security practices to ensure confidentiality and integrity.

• **Requirements:**

- All data transfers must be encrypted using **TLS 1.3**.
- Sensitive data (passwords, health metrics) must be stored using **AES-256 encryption**.
- System must comply with relevant Indian data protection standards (e.g., **DPDP Act 2023**).
- Multi-factor authentication should be available for enhanced protection.

---

3. Usability

• **Description:**
The platform should be intuitive, user-friendly, and comfortable for users of all ages and technical backgrounds.

• **Requirements:**

- Users should navigate the app easily with minimal instruction.
- Interface design must maintain consistency in layout, icons, and interaction.
- Accessibility support (voice input, large icons, color contrast) should meet **WCAG 2.1** standards.

---

4. Compatibility

• **Description:**
The platform should function seamlessly across a wide variety of devices and browsers.

• **Requirements:**

- Must be fully responsive on smartphones, tablets, and desktops.
- Should support major browsers (Chrome, Firefox, Safari, Edge).
- Android (11+) and iOS (14+) compatibility required for mobile app versions.

---

5. Reliability

• **Description:**
The platform must operate consistently and be available whenever users need to track or manage wellness data.

• **Requirements:**

- System uptime should be at least **99.5%** excluding scheduled maintenance.
- Automatic backup and recovery should prevent data loss during failures.
- AI and analytics modules should continue functioning even under heavy load.

---

6. Scalability

• **Description:**
The system should be capable of expanding in response to increasing users, data volume, and wellness features.

• **Requirements:**

- Architecture should allow scaling to support **2× user growth annually**.
- Database and API services should handle large volumes of logs, health entries, and workout data.
- AI models should support incremental retraining without downtime.

---

7. Maintainability

• **Description:**
The platform codebase and infrastructure should be easy to manage, update, and debug over time.

• **Requirements:**

- Code must follow clean coding standards with proper documentation.
- Updates, bug fixes, and new features should be deployable with **minimal downtime** using CI/CD pipelines.
- System logs and admin dashboards should help quickly identify issues.

2.5.4 Performance Requirements

1. Response Time

• **Requirement:**
The system should respond to user actions (such as logging meals, starting workouts, loading dashboards, or chatting with the AI Coach) within **2 seconds** under normal operating conditions.

• **Justification:**
Faster response times enhance user experience, reduce frustration, and support continuous engagement with wellness activities.

---

2. Throughput

• **Requirement:**
The system must support at least **5,000 concurrent users** without noticeable degradation in performance.

• **Justification:**
WellnexUS may experience peak traffic during morning and evening wellness hours, requiring stable multi-user handling.

---

3. Load Handling

• **Requirement:**
The application must maintain optimal performance even when multiple users are simultaneously logging meals, tracking workouts, and using AI chat features.

• **Justification:**
High load tolerance prevents crashes or slowdowns during peak fitness or challenge periods.

---

4. Caching Mechanism

• **Requirement:**
Implement caching for frequently accessed data such as food items, workout templates, and user dashboards to minimize server requests.

• **Justification:**
Caching reduces database load and significantly improves page load speed.

---

5. AI Processing Time

• **Requirement:**
AI-generated responses (chatbot replies, recommendations, analytics insights) should be delivered within **3 seconds**.

• **Justification:**
Real-time interactions are crucial for wellness guidance, user satisfaction, and maintaining conversation flow.

---

6. Data Synchronization Rate

• **Requirement:**
User activity data (steps, heart rate, calories burned) should sync with connected devices every **10–20 seconds** when active.

• **Justification:**
Accurate and timely synchronization is essential for real-time wellness tracking and analytics.

---

7. Report Generation Speed

• **Requirement:**
Weekly and monthly wellness reports (PDF or in-app analytics) should generate within **10 seconds**.

• **Justification:**
Efficient report generation improves usability and encourages regular progress review by users and coaches.

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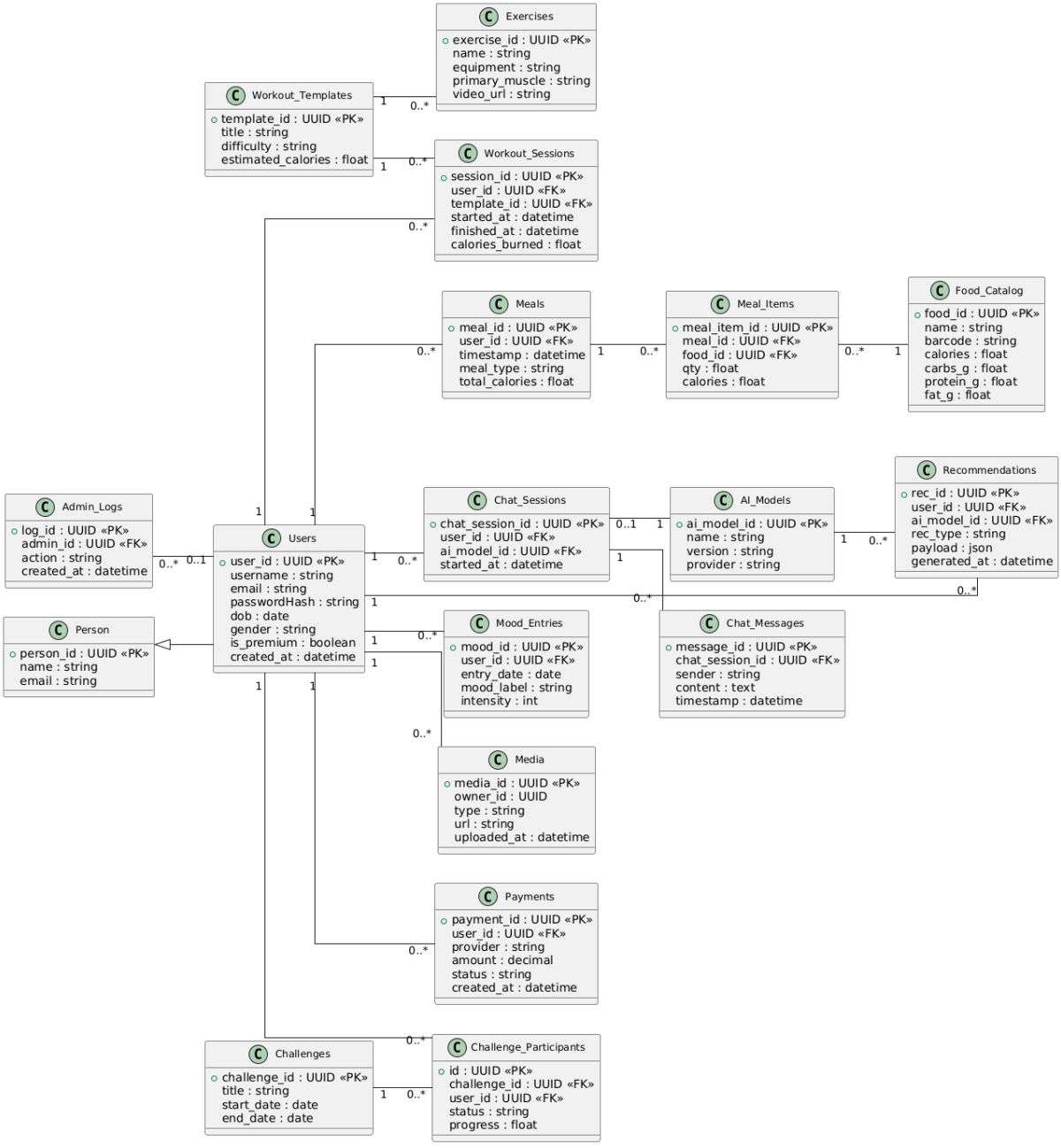
8. Server Utilization Efficiency

• **Requirement:**
During peak hours, CPU usage should not exceed **80%**, and memory usage should remain under **85%** of allocated resources.

• **Justification:**
Maintaining balanced resource utilization prevents unexpected downtime and ensures smooth system performance.

2.5.5 ER Diagram

The ER diagram for the WellnexUS project illustrates the relationships among core entities such as User, Meals, Workout_Sessions, Mood_Entries, Chat_Sessions, and AI_Models. Each user can log multiple meals connected to detailed food data in the Food_Catalog, and track workouts through Workout_Templates and Exercises. Emotional and mental wellness are managed via Mood_Entries and interactive Chat_Sessions, which link to AI_Models generating personalized Recommendations. Supporting entities like Payments, Challenges, Media, and Admin_Logs maintain premium features, social engagement, and system control. These relationships collectively ensure an integrated approach to health, fitness, and wellness management within a unified platform.



2.6 User Stories / Story Cards

#101 User Registration
As a new user, I want to create an account so that I can access all WellnexUS wellness features.
Confirmation:
1. Success – display message: "Welcome to WellnexUS! Let's begin your wellness journey."
2. Failure – display message: a. "Invalid email or password." b. "Network error – please try again." c. "Server not responding."

#102 Health Profile Setup
As a new user, I want to complete my health profile so that I can get personalized wellness recommendations.
Confirmation:
1. Success – display message: "Profile setup complete! Personalized insights are ready."
2. Failure – display message: a. "Incomplete information – please fill all required fields." b. "Unable to save profile. Check your internet connection."

#201 Food Logging
As a user, I want to log my meals easily so that I can monitor my daily nutrition.
Confirmation:
1. Success – display message: "Meal logged successfully. Calories updated."
2. Failure – display message: a. "Invalid food entry." b. "Unable to connect to food database." c. "Network error – please try again later."

#301 Workout Sessions
As a user, I want to follow guided workout sessions so that I can perform exercises correctly.
Confirmation:
1. Success – display message: "Workout session completed! Great job!"
2. Failure – display message: a. "Video unavailable or failed to load." b. "Unable to record progress." c. "Internet connection error."

#401 Mood Tracking
As a user, I want to record my mood daily so that I can understand my emotional patterns.
Confirmation:
1. Success – display message: "Mood recorded successfully. View trends in dashboard."
2. Failure – display message: a. "Please select a mood before saving." b. "Unable to sync data with server."

#402 AI Wellness Chat
As a user, I want to chat with the AI wellness coach so that I can receive personalized guidance.
Confirmation:
1. Success – display message: "Conversation saved. AI insights generated successfully."
2. Failure – display message: a. "AI service temporarily unavailable." b. "Failed to fetch response – check connection."

#501 Health Dashboard
As a user, I want to view my daily progress summary so that I can track my fitness, nutrition, and wellness in one place.
Confirmation:
1. Success – display message: "Dashboard loaded successfully. Here's your progress!"
2. Failure – display message: a. "Data unavailable – please refresh." b. "Server not responding."

#601 Wellness Challenges
As a user, I want to participate in community wellness challenges so that I can stay motivated with others.
Confirmation:
1. Success – display message: "Challenge joined successfully! Let's stay consistent."
2. Failure – display message: a. "Challenge registration failed." b. "Network error – please try again later."

Conclusion

This Software Requirements Specification document outlines the complete foundation for the development of **WellnexUS**, a holistic wellbeing platform that unifies physical fitness, nutrition management, and mental wellness into a single application. The document defines the system's objectives, scope, functional and non-functional requirements, user stories, and system models to ensure clarity and consistency across all stages of development.

The **WellnexUS** platform is designed to promote healthy living through personalized recommendations, AI-driven insights, and an engaging user experience. By integrating wellness tracking, community participation, and real-time analytics, the system offers users a comprehensive view of their overall health and progress.

This SRS provides a clear roadmap for developers, designers, and stakeholders, ensuring the successful transition from analysis to design and implementation. With its detailed requirements and structured specifications, the document ensures that all aspects of system functionality, performance, and usability are well-defined and achievable.